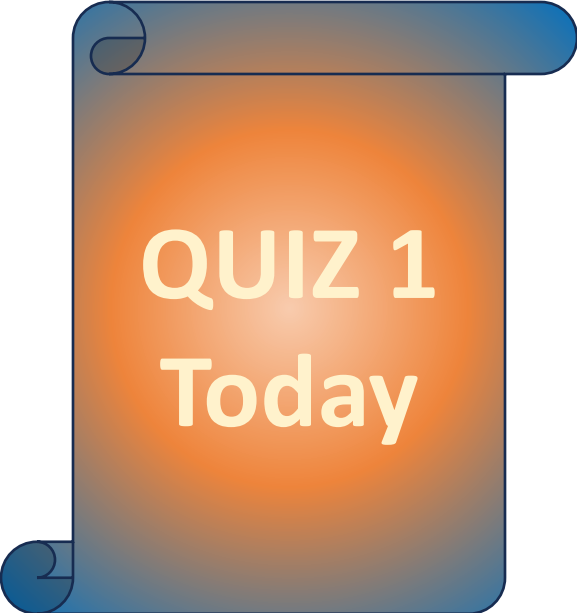


Lecture 5

EE 421 / CS 425

Digital System Design

Fall 2026
Shahid Masud



QUIZ 1
Today

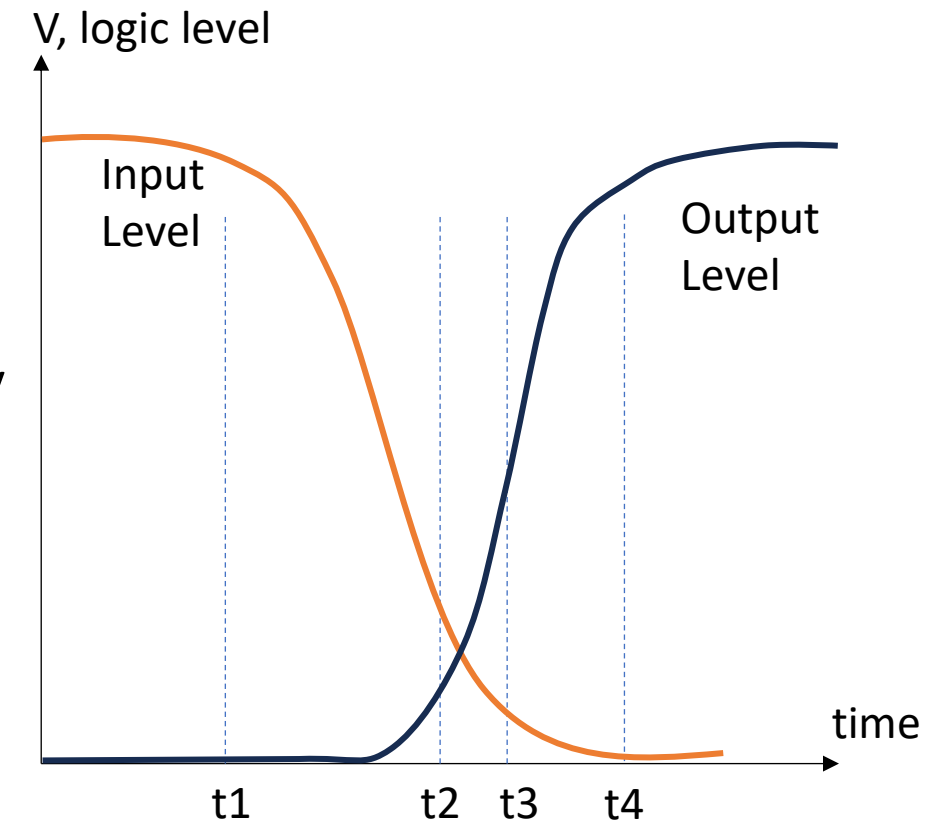
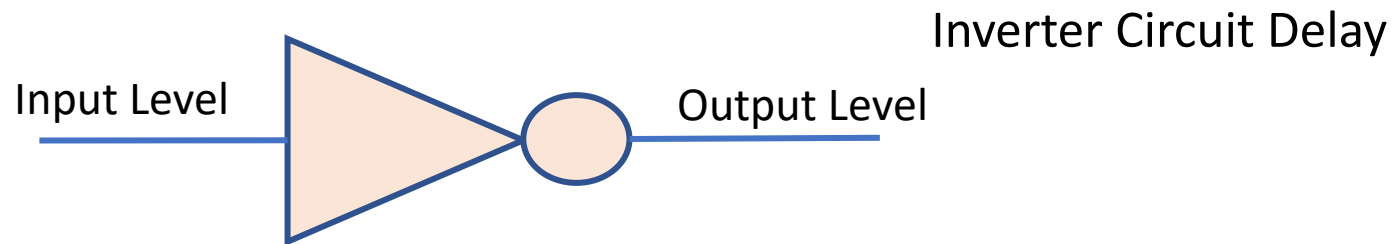
Topics

- Pulse Circuits
- Timing Related Effects in Combinational Circuits
- Glitches
- Hazards
- Identify and Remove Glitches and Hazards
- If we have time, we may start
 - Review of Sequential Digital Logic Circuits
 - Basics of Latches
 - Basics of Flipflops
- QUIZ 1 TODAY

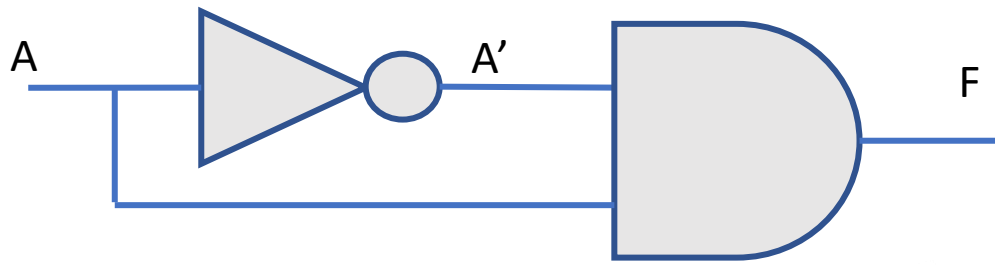
Timing Waveforms

Every physical circuit has some processing delay; once the input has settled, **the correct output appears after some time.**

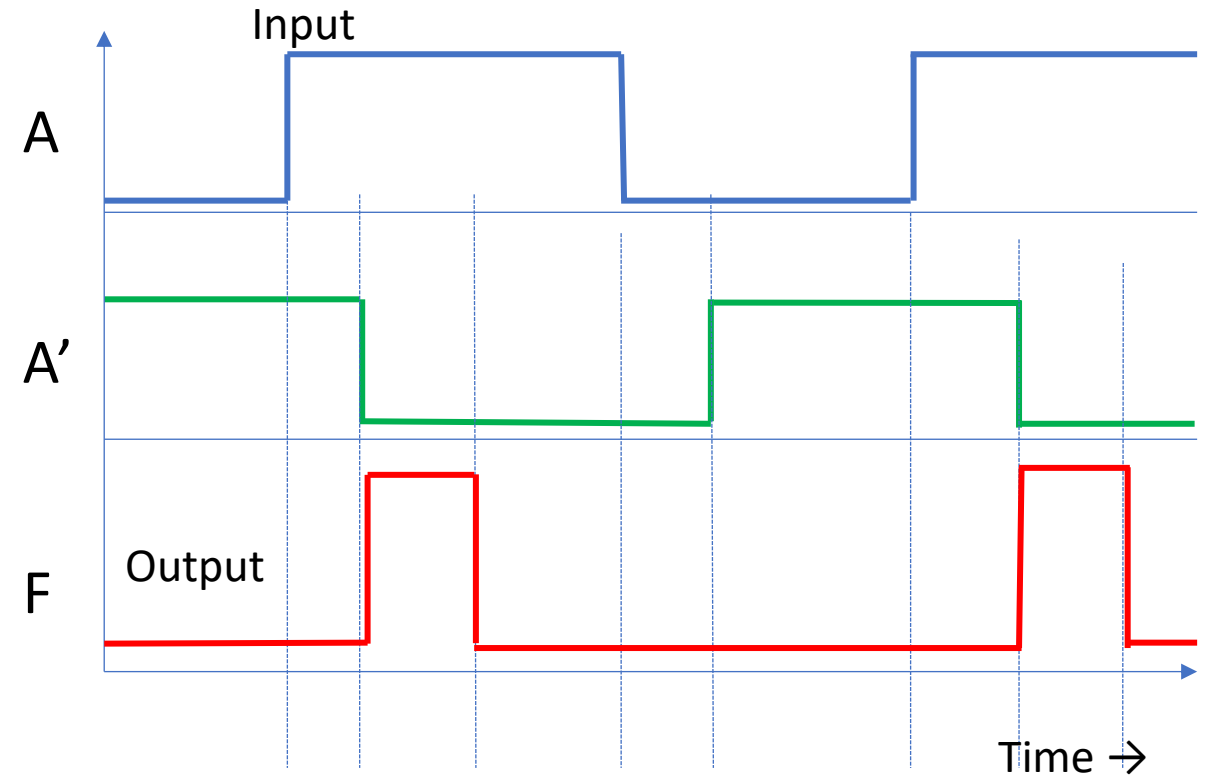
Propagation delay from a gate is defined by:
 t_{plh} = time delay in output going from low to high
 t_{phl} = time delay in output going from high to low



A simple pulse circuit



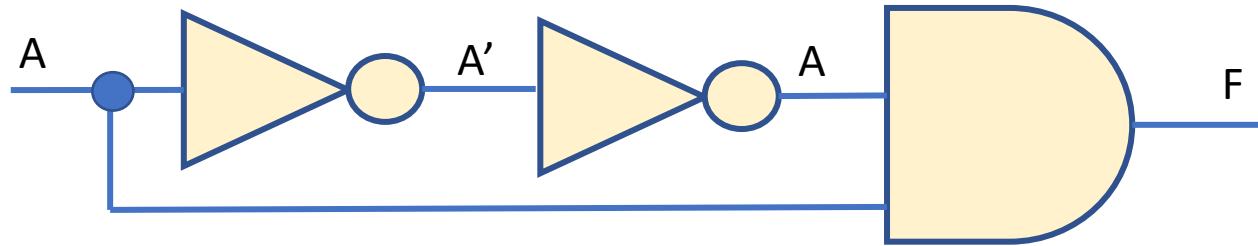
Each gate has Propagation Delay



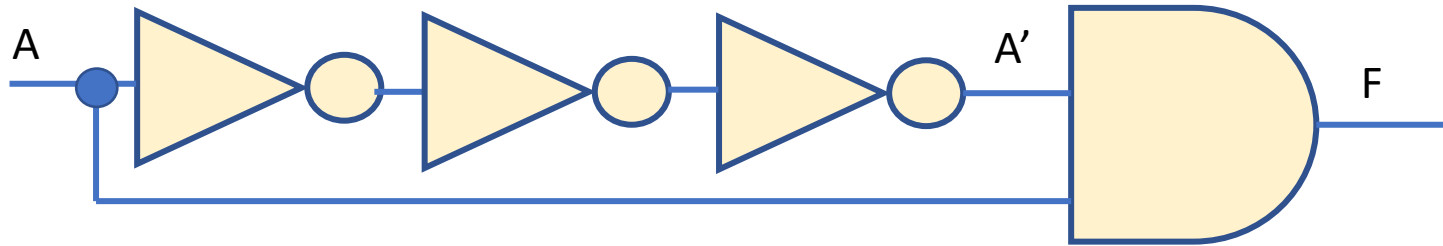
Question: What will be the Output if there are two inverters in series?

Question: What will be the Output if there are three inverters in series?

Two and Three Inverters in simple pulse circuit

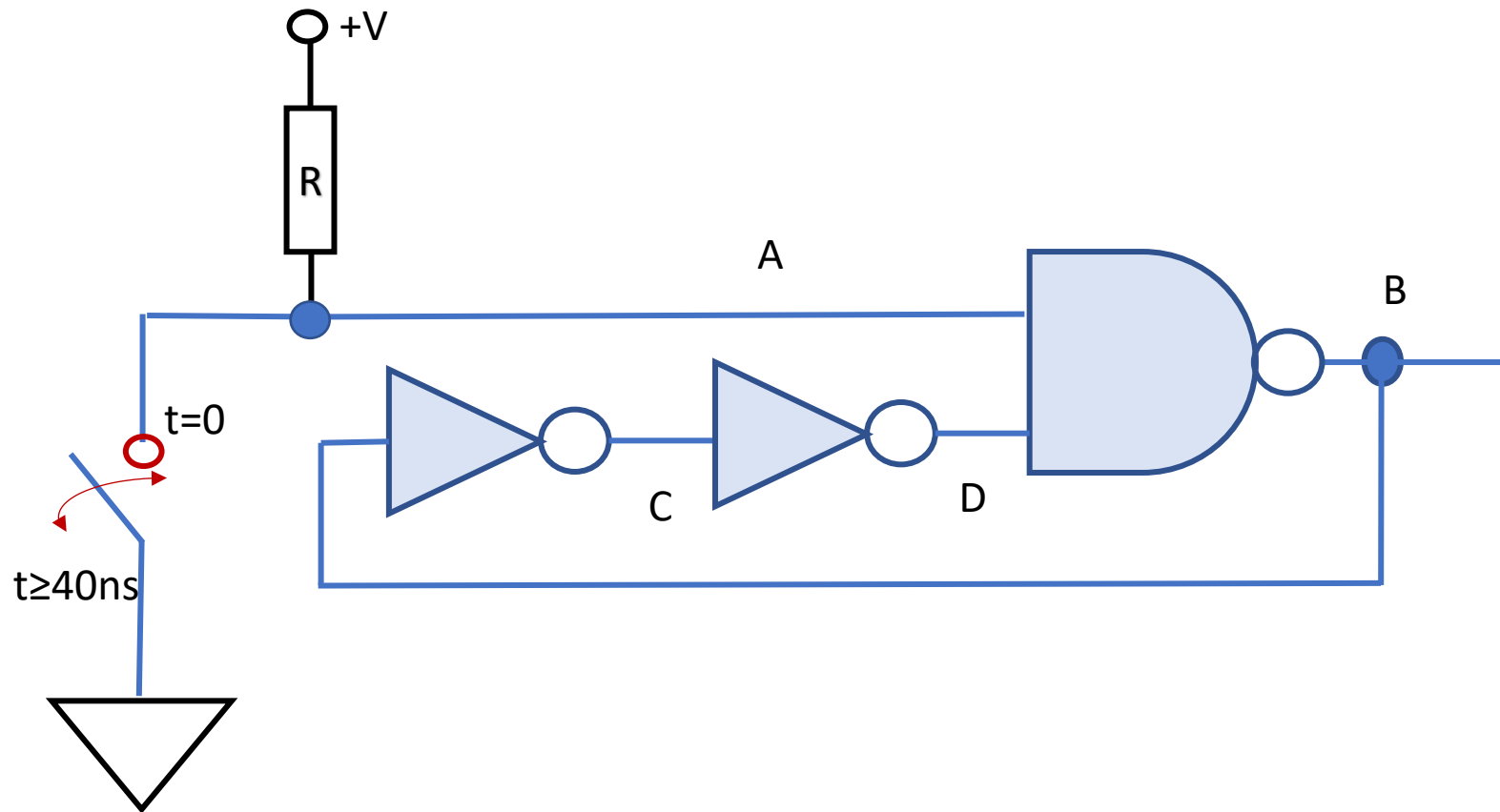


For even number of Inverters:
Check input A transitions from
0 to 1 and from 1 to 0
Pulse behaviour is not observed
(Hint: Check whether F is like A?)



For Odd number of Inverters:
At 0 to 1 transitions at A, a pulse will
appear at F. The pulse width is proportional
to number of inverters as their propagation
delay adds up

A Pulse Shaper Circuit



Case 1:

Assume propagation delay of gates as 10ns

Switch is closed at $t=0\text{ns}$, thus:

A goes to '0' at time $t=0\text{ns}$

B goes to '1' at time $t=10\text{ns}$

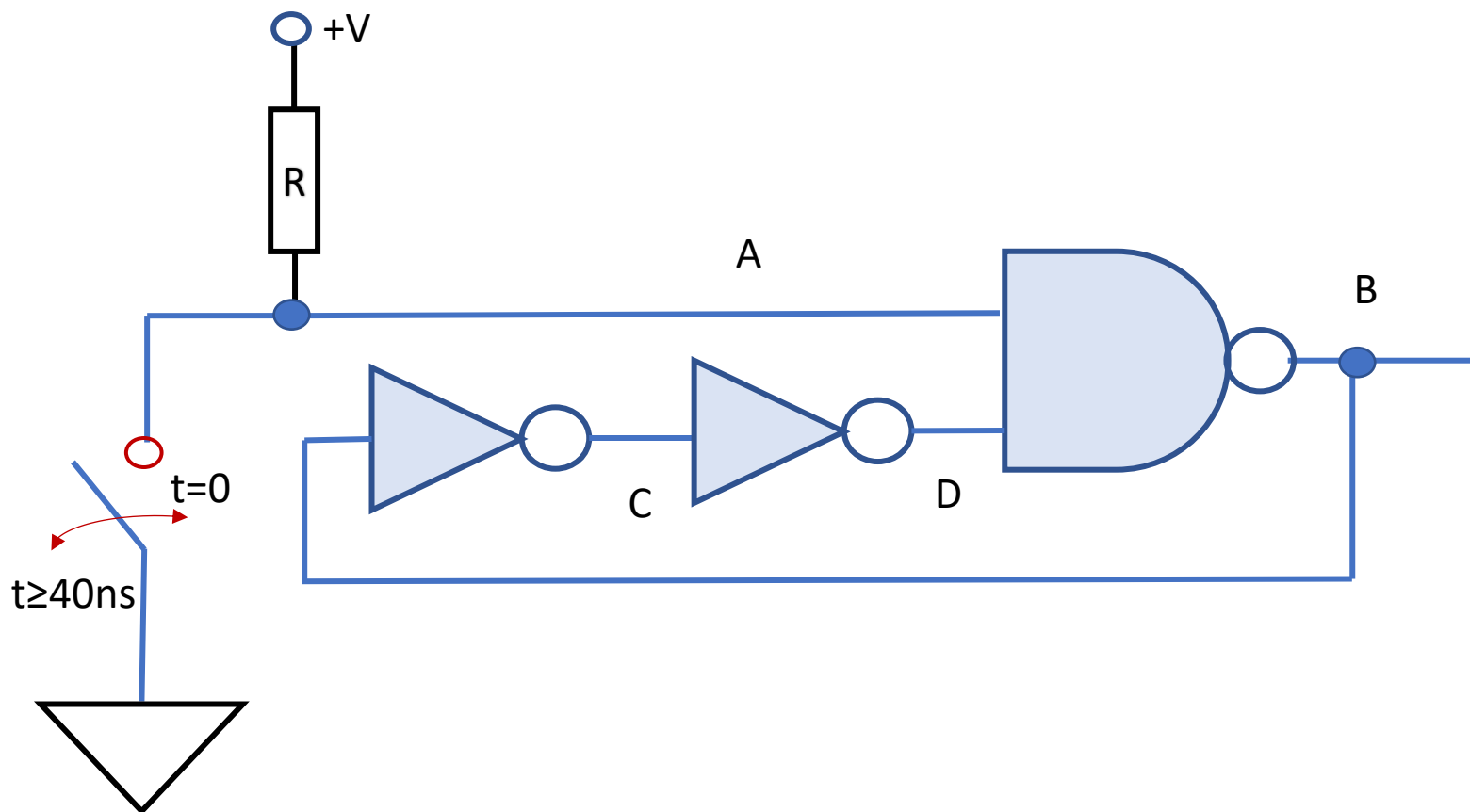
C goes to '0' at time $t=20\text{ns}$

D goes to '1' at time $t=30\text{ns}$

Since $D='1'$, $A='0'$, hence:

Output $B='0'$ always for $A='0'$ in steady state

Pulse Shaper Circuit – Case 2



Case 2:

Assume propagation delay of gates as 10ns

Output B was in steady state = '0' while Switch was closed at $t=0$

Now Switch is opened at $t=40\text{ns}$

So A goes to '1' at $t=40\text{ns}$

B goes to '0' after 10ns at $t=50\text{ns}$

C goes to '1' at $t=60\text{ns}$

D goes to '0' at $t=70\text{ns}$

Now $D='0'$ and $A='1'$, hence $B='1'$ at $t=80\text{ns}$

C goes to '0' at $t=90\text{ns}$

D goes to '1' at $t=100\text{ns}$

Since $A='1'$ and $D='1'$ so $B='0'$ at $t=110\text{ns}$

Time Period of this oscillator is determined by propagation delay of gates

Same behavior can be obtained from a NOR gate circuit with some changes

The cycle will start repeating automatically

We get an **OSCILLATOR**

Glitches and Hazards

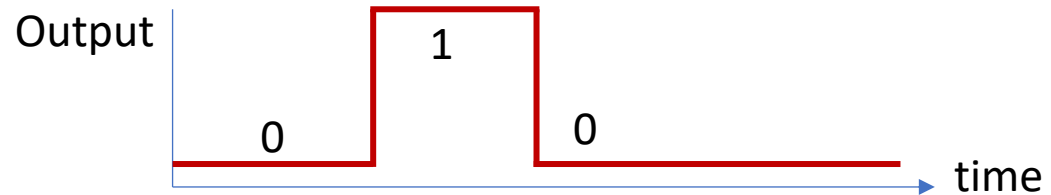
Glitch: An Unwanted Pulse at the Output of a Combinational Logic Circuit. Glitch is dependent upon Inputs and how the circuit is configured

Hazard: A Combinational Logic Circuit with a potential to produce Glitches is called a Hazard.

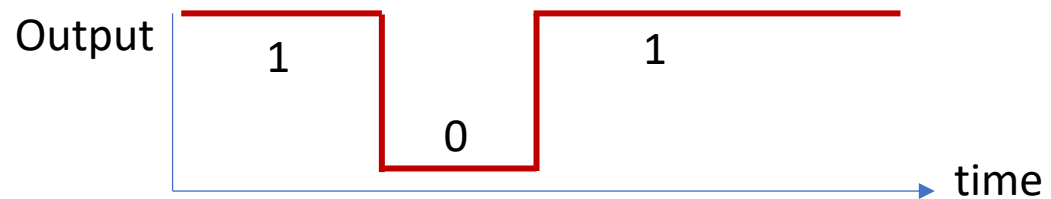
Possible Reasons:

1. Using Un-Stable Input Signals
2. Not Considering Propagation Delays in high-speed designs
3. Using Asynchronous Inputs

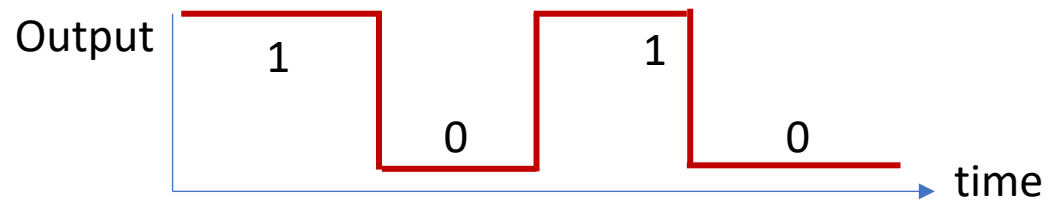
Kinds of Hazards



Input Changes cause output to change from '0' to '1' to '0'
STATIC ZERO HAZARD

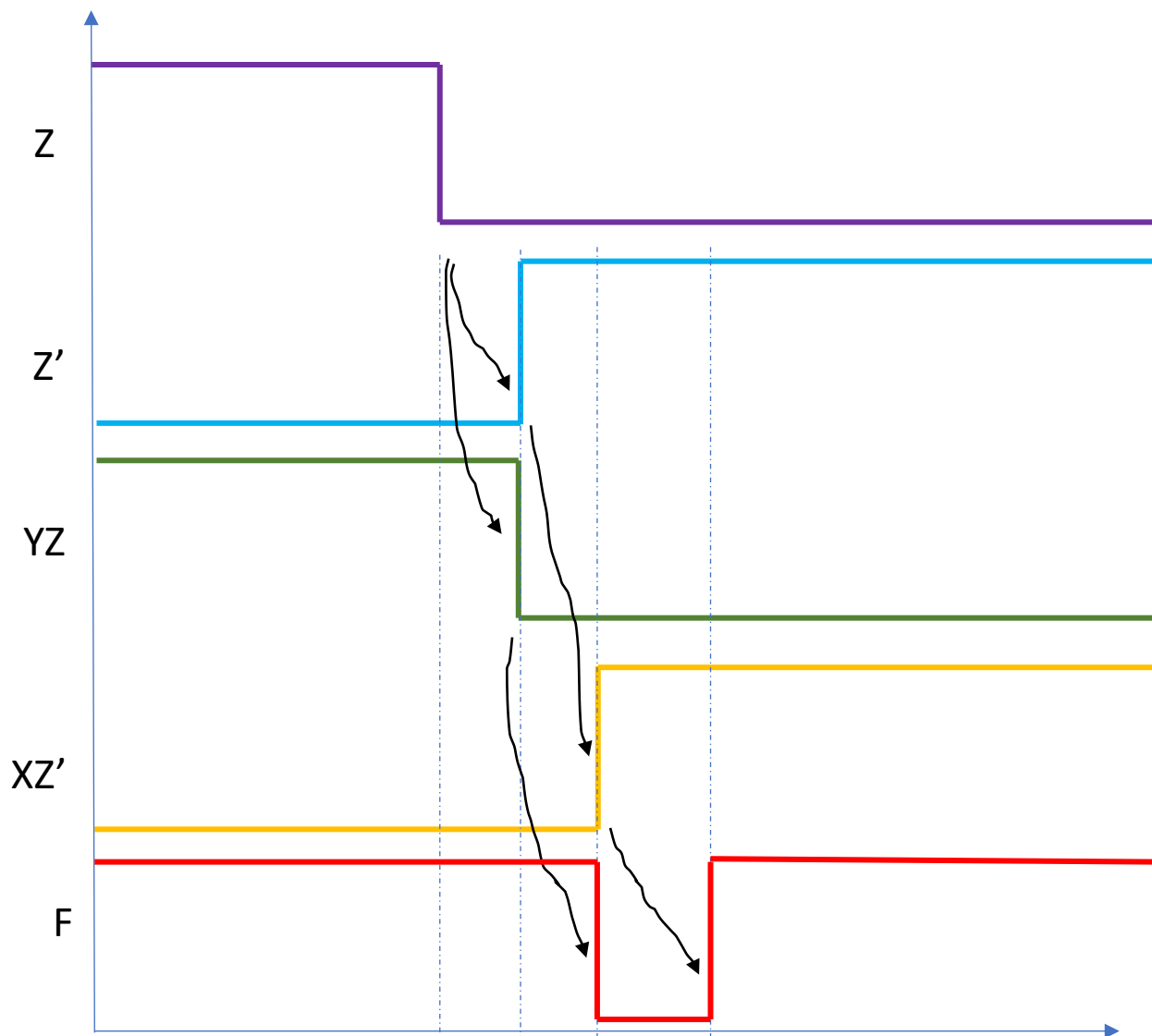
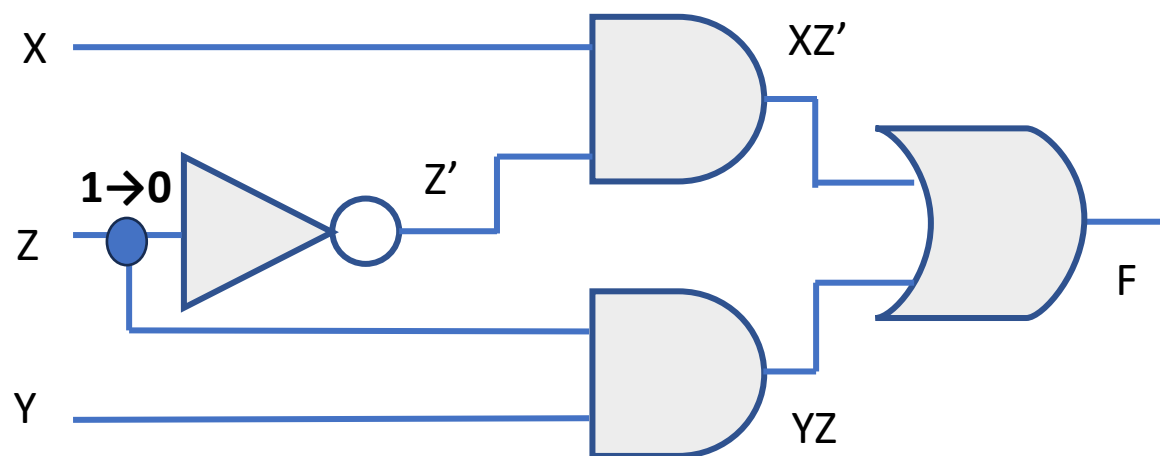


Input Changes cause output to change from '1' to '0' to '1'
STATIC ONE HAZARD



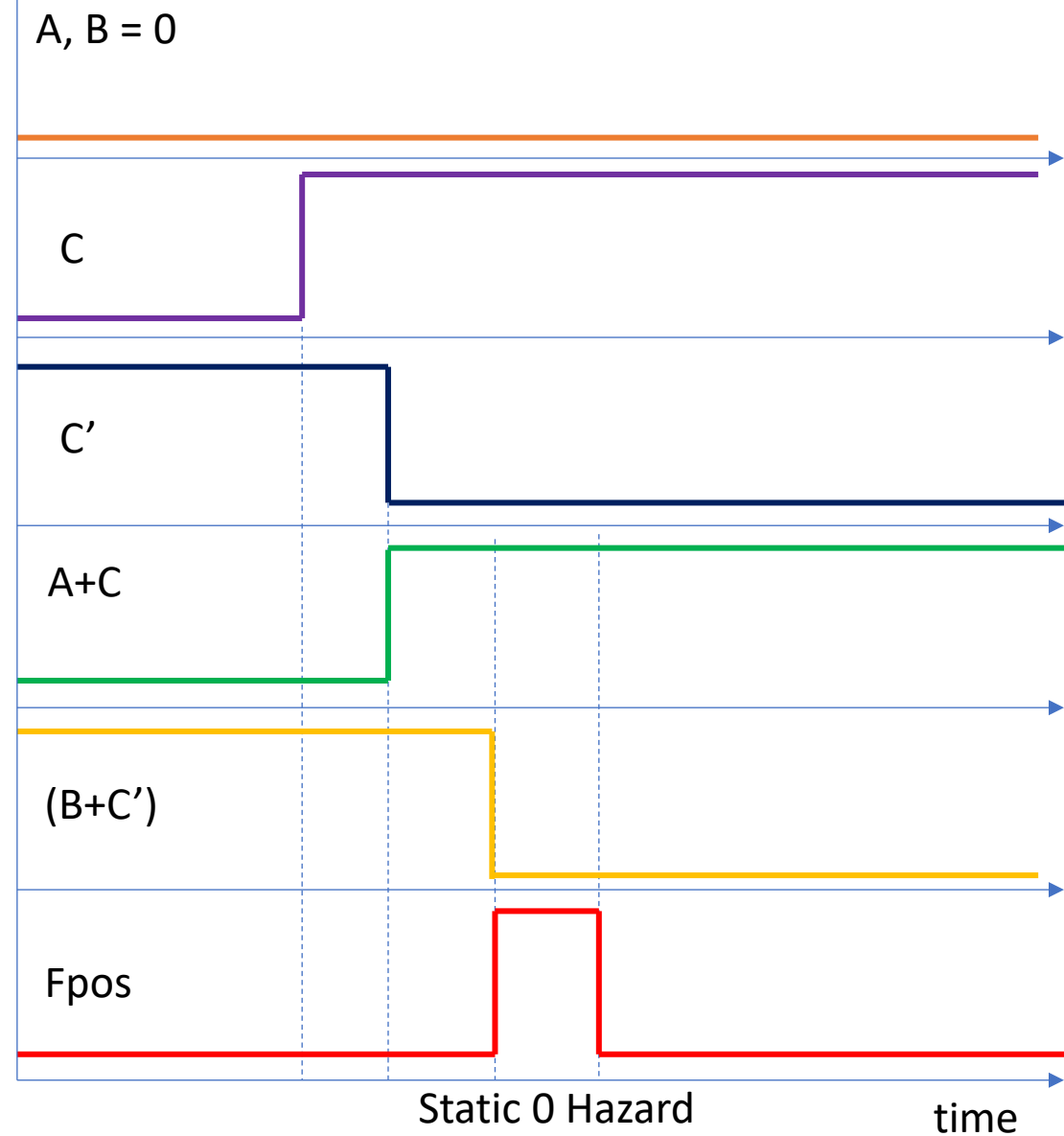
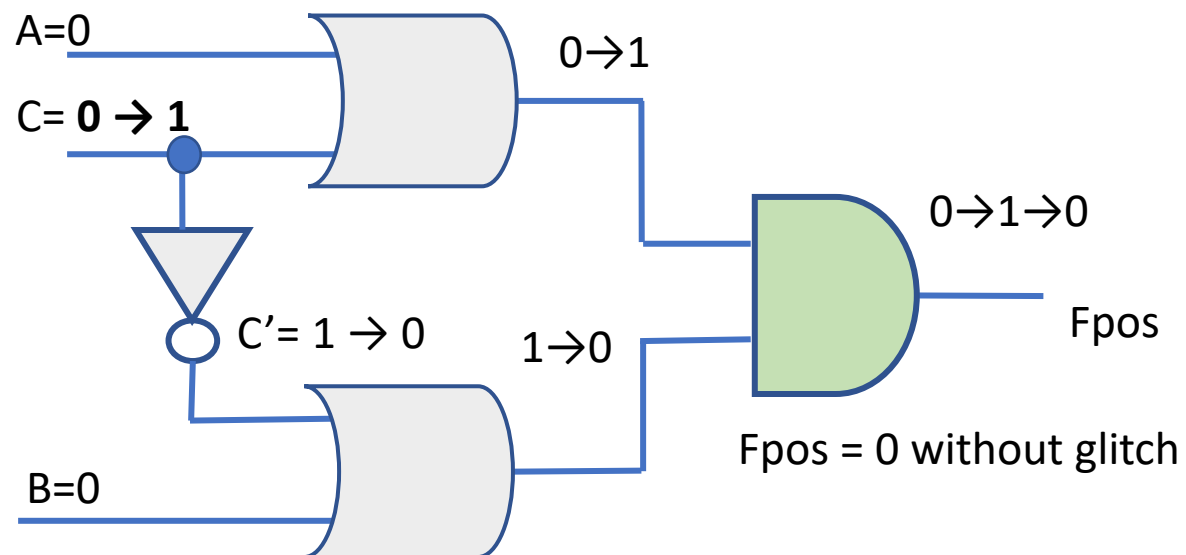
Input Changes cause two or more changes from '1' to '0' to '1' to '0', Or changes from '0' to '1' to '0' to '1'
DYNAMIC HAZARD

Circuit with Static 1 Hazard

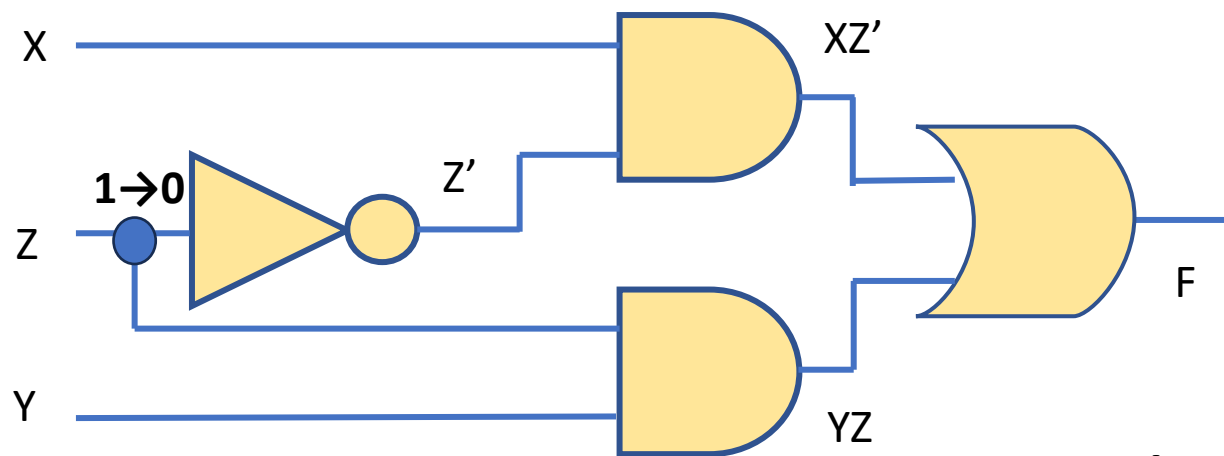


Static 1 Hazard Observed

Example of Static 0 Hazard



K Maps to Identify Hazards



$$F = XZ' + YZ$$

Z, XY	00	01	11	10
0	0	0	1	1
1	0	1	1	0

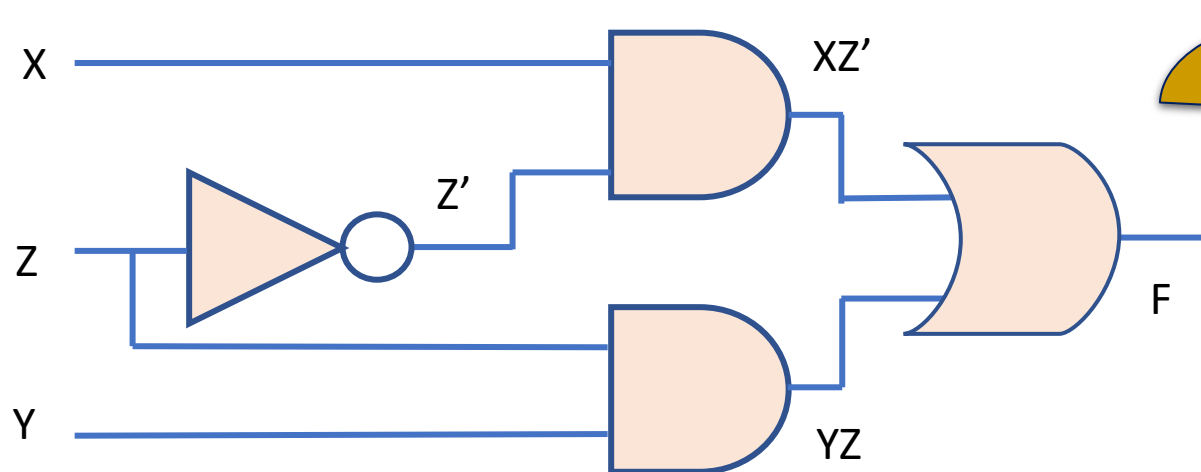
YZ

XZ'

How to Know about Hazards?

- A properly designed two level SOP (AND-OR) circuit has no Static-0 Hazard
- This type of SOP circuits can have Static-1 Hazards
- Look at the K-Map again and we notice that there is no single product term that covers the inputs $XYZ=111$ and $XYZ=110$. Thus glitch to 0 is possible if the AND gate covering the other combination goes to 1.
- **To Eliminate Such Hazards:**
- Include an extra product term (AND gate) to cover the hazard input. This is where the adjacent 1s exist in K-Maps but are not covered. This is called 'Adding a Consensus Term'

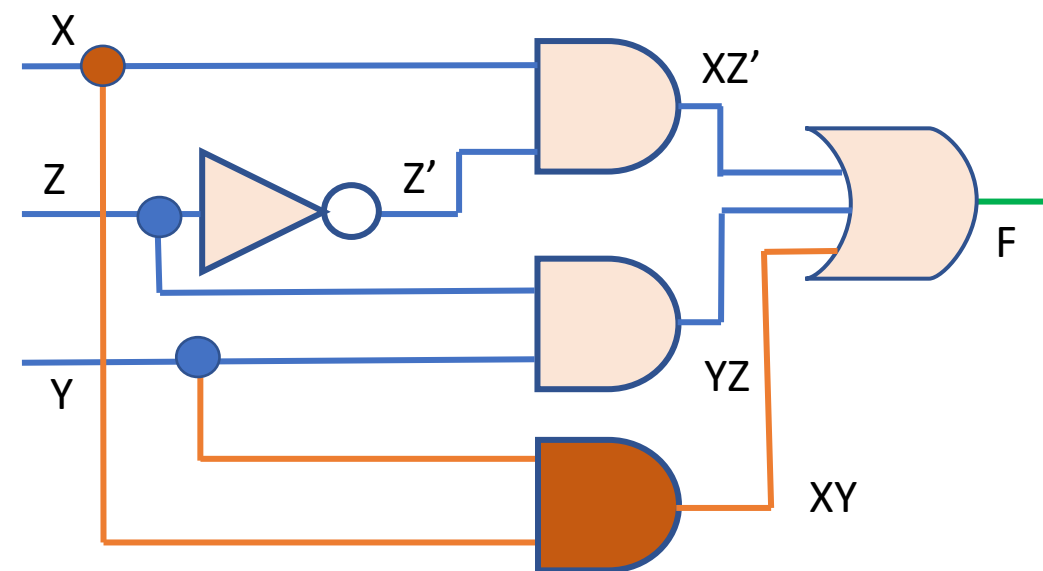
Hazard Free two level SOP Circuit



Truth Table for $F = XZ' + YZ$:

Z, XY	00	01	11	10
0	0	0	1	1
1	0	1	1	0

Annotations: The 1s in the 11 and 10 columns are grouped by a blue box labeled XZ' . The 1s in the 01 and 11 rows are grouped by a green oval labeled YZ . The intersection of these groups (the 11 cell) is labeled "Consensus Term XY".



F including Consensus Term = $XY + YZ + XZ'$

**Final Hazard Free Circuit
Including Extra Consensus Circuit**

The Adjacent 1s in K-Map are now Covered

Using K-Map to Identify Static 1 Hazard in 4 input variables

AB, CD	00	01	11	10
00	0	1	1	0
01	1	1	0	0
11	1	1	1	1
10	0	0	1	1

Given K-Map with Minimal Grouping

Identify the Hazards

Adjacent 1s that are not in the same group

K-Map Groups to Eliminate Static 1 Hazard

YZ, WX	00	01	10	11
00	0	1	1	0
01	1	1	0	0
11	1	1	1	1
10	0	0	1	1

$XY'Z'$ (points to cell 01, 10)
 $W'Z$ (points to cell 01, 10)
 WY (points to cell 11, 10)

Minimal Expression:
 $F = XY'Z' + W'Z + WY$



AB, CD	00	01	10	11
00	0	1	1	0
01	1	1	0	0
11	1	1	1	1
10	0	0	1	1

$W'XY'$ (points to cell 01, 00)
 WXZ' (points to cell 10, 00)
 YZ (points to cell 11, 11)

Hazard Free Expression:
 $F = XY'Z' + W'Z + WY + W'XY' + YZ + WXZ'$

K-Map Grouping to Eliminate Static 0 Hazard

POS = Grouping of Zeros

AB, CD	00	01	10	11
00	1	1	0	0
01	1	1	1	1
11	0	0	1	1
10	0	0	0	0

$(A' + C)$

$(B + C')$

Minimal Mapping without Hazard Consideration

$$F = (A' + C) (B + C')$$

AB, CD	00	01	10	11
00	1	1	0	0
01	1	1	1	1
11	0	0	1	1
10	0	0	0	0

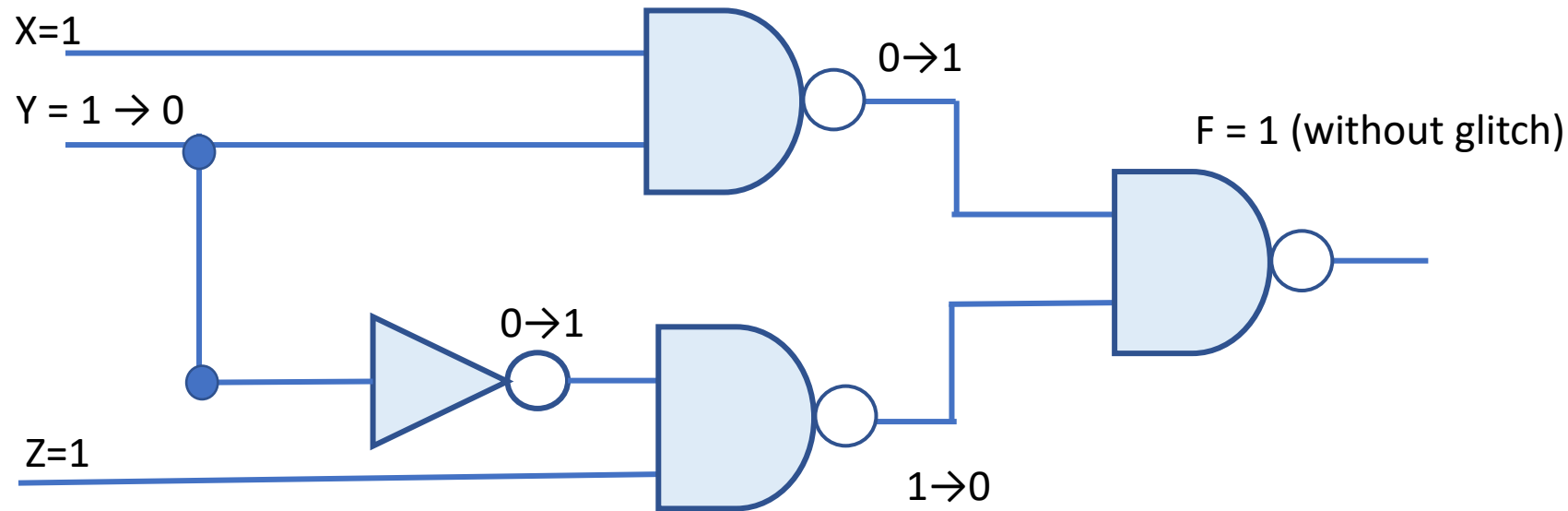
$(A' + B)$

Additional Cover for Static 0 Hazard prevention

$$F = (A' + C) (B + C') (A' + B)$$

Now free from Static 0 Hazard

Example of Static 0 Hazard



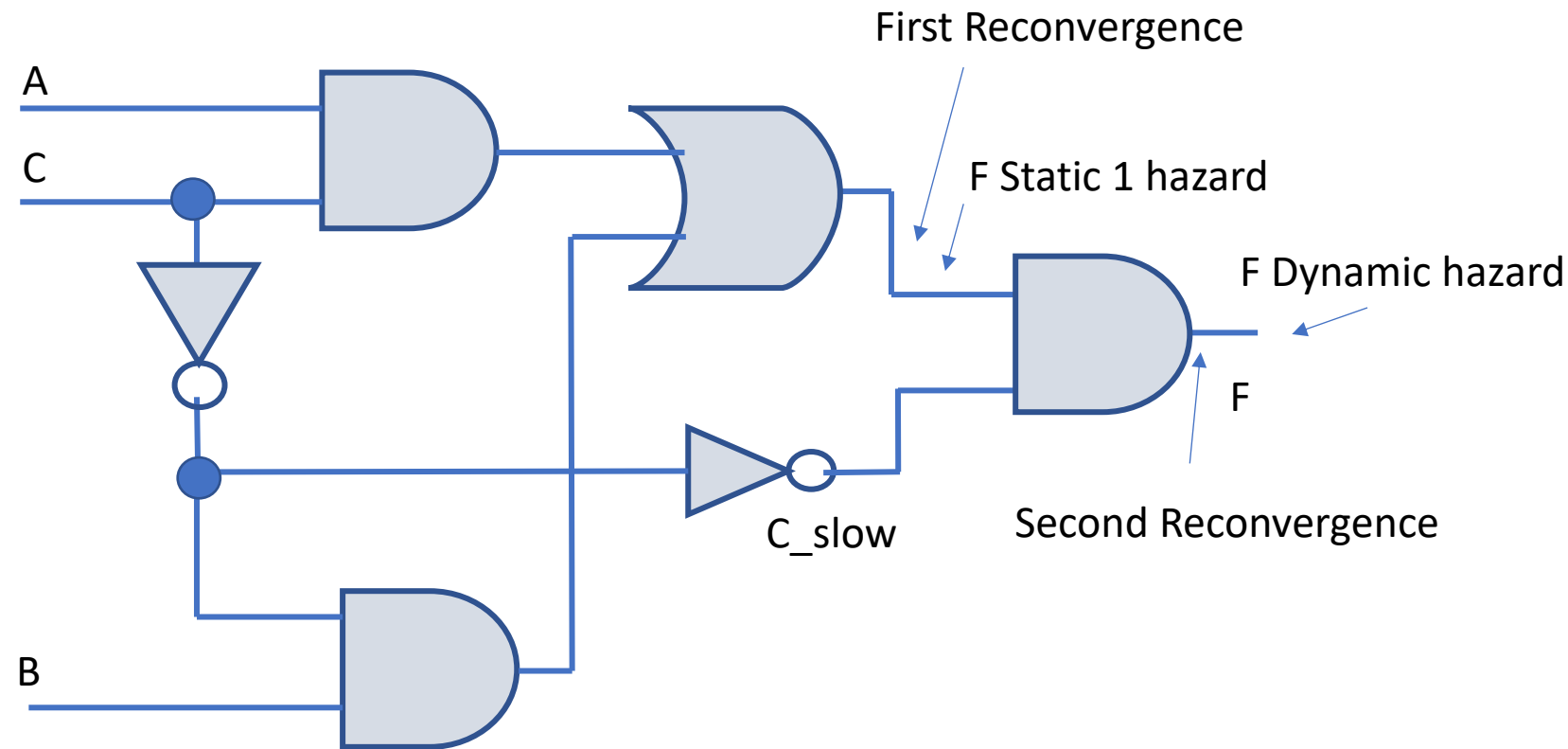
$$F=(X+Y').(Y+Z)$$

When Y changes from 1 to 0, both inputs of Nand Gate F may be equal to 1, Causing the output to momentarily go to 0 when it should have stayed at 1.

Dynamic Hazards

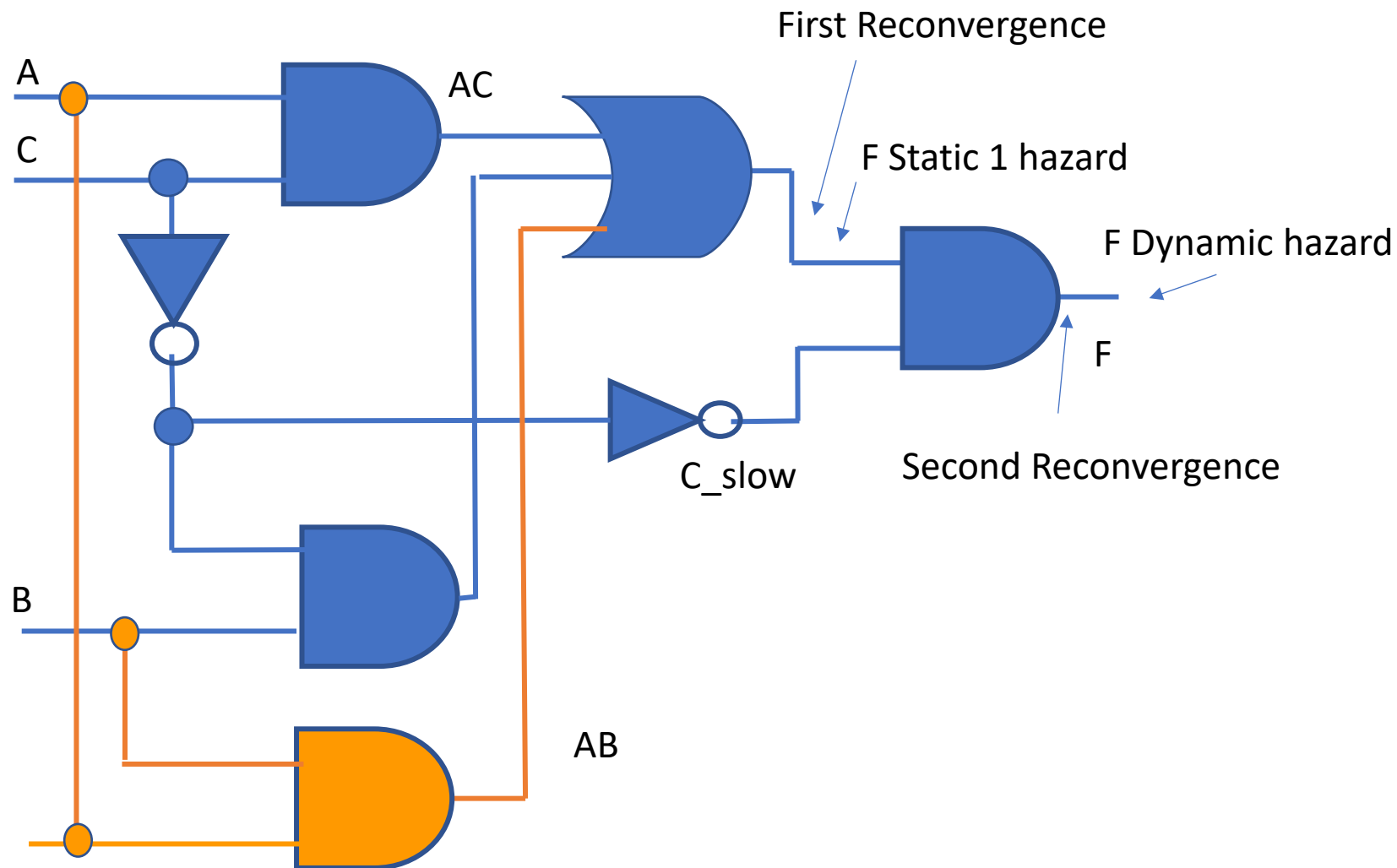
- Dynamic Hazard is the possibility of an output changing more than once as the result of a single input transition.
- These can occur if there are multiple paths with different delays from the input to the output
- Dynamic Hazards do not occur in properly designed two level AND-OR or OR-AND Circuits.
- Thus remove all Static-1 and Static-0 Hazards in the two-level SOP or POS design and the possibility of Dynamic Hazards is minimized.

Circuit containing Dynamic Hazard



Solution: Make K Map and add redundant cover for C

Redundant Circuit to remove Dynamic Hazard

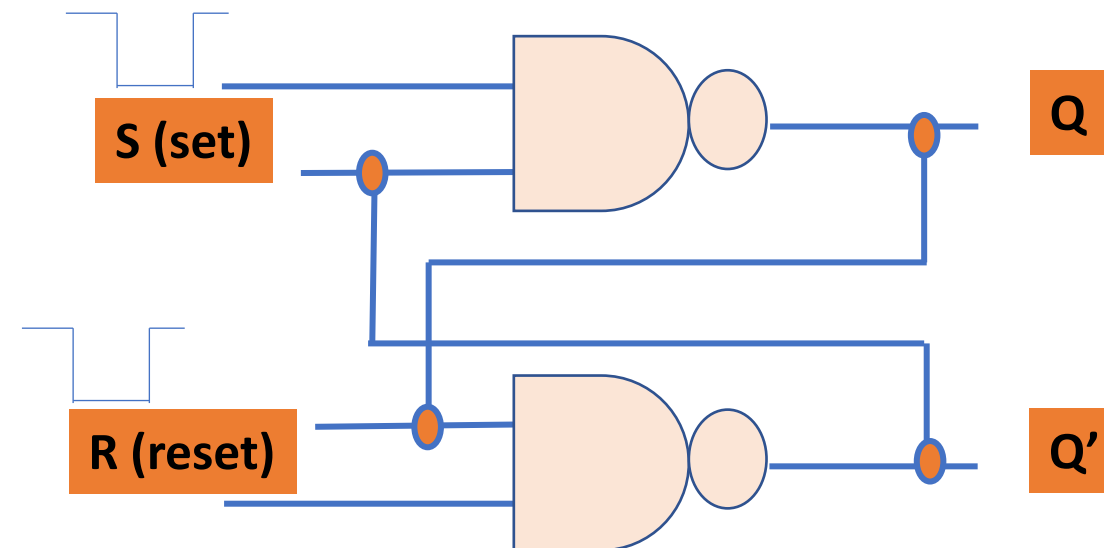
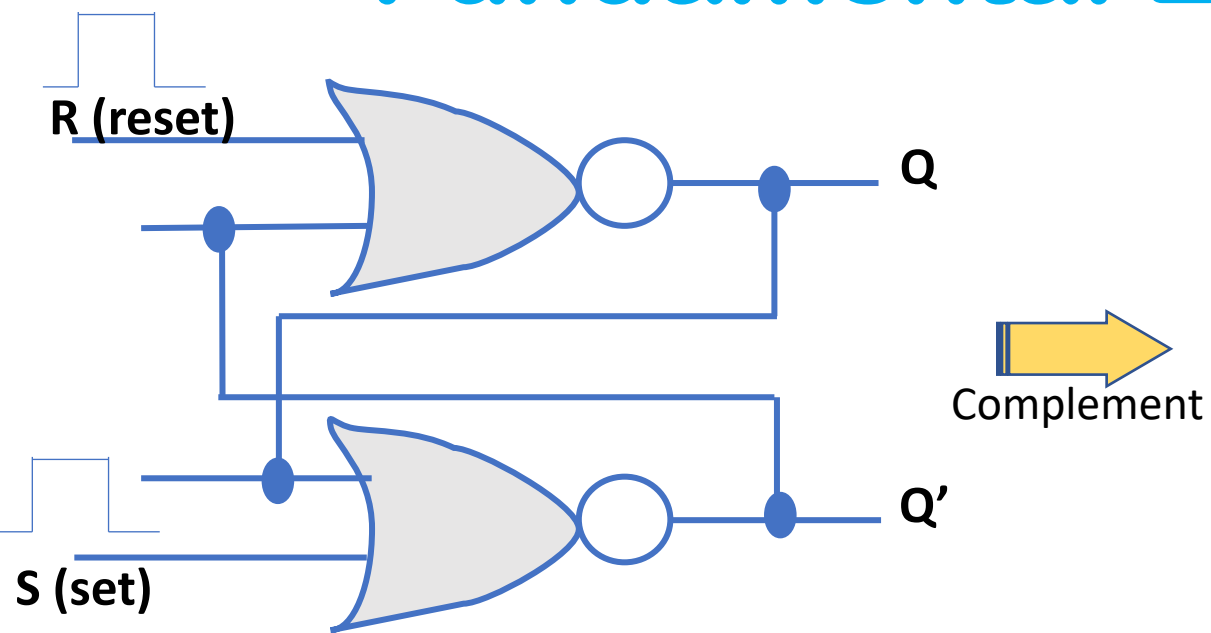


- **We may start review of Sequential Elements and Circuits**

Latches and Flipflops - Definitions

- **Latches:** Level Sensitive storage elements; that can transition at any point in time
- **Flipflops:** Edge Sensitive storage elements, that transition at clock edges, either rising or falling
- **Asynchronous Sequential Logic:** Based on latches and un-clocked transitions
- **Synchronous Sequential Logic:** Based on flipflops and only clocked transitions

Fundamental Element – SR Latch



SR Latch Using 2 Input NOR Gate – Function Table

S (Set)	R (Reset)	Q	Condition
0	0	Q (=1 after S=1, R=0)	Hold
0	0	Q (=0 after S=0, R=1)	Hold
0	1	0	Reset
1	0	1	Set
1	1	0	Not Allowed

RACE
Condition

SR Latch Using 2 Input NAND Gate – Function Table

S	R	Q	Condition
0	0	1	Not Allowed
0	1	1	Set
1	0	0	Reset
1	1	Q (=0 after S=1, R=0)	Hold
1	1	Q (=1 after S=0, R=1)	Hold

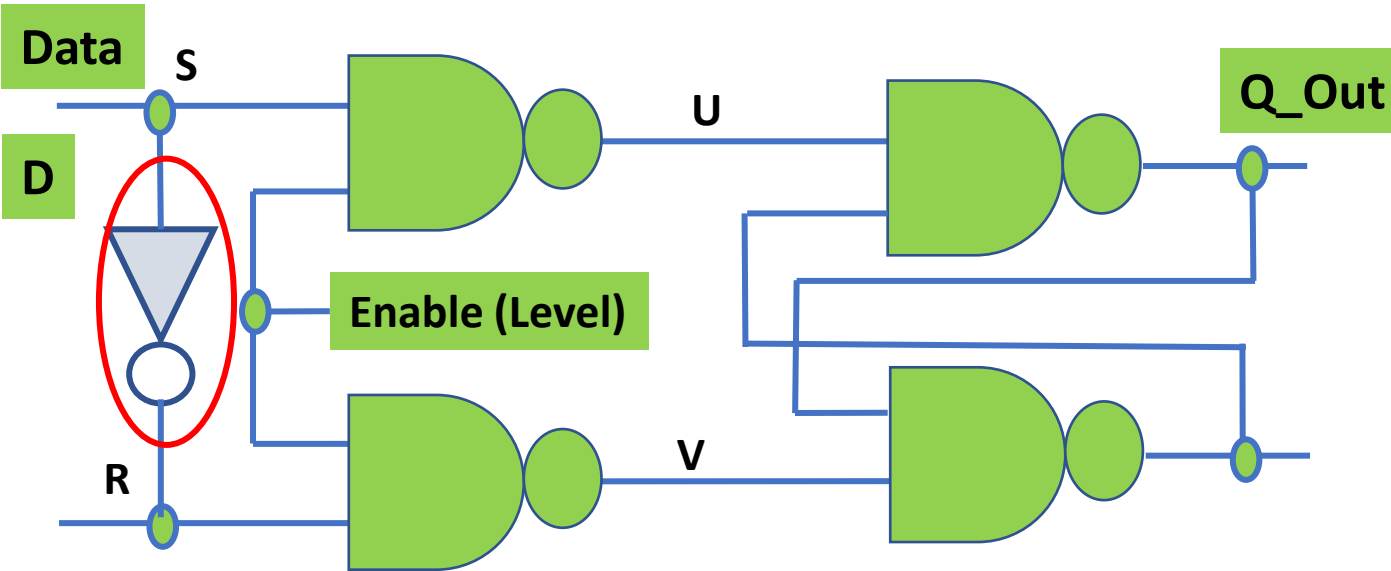
RACE
Condition

Transparent Latch

- An additional 'Enable' input is provided
- Definition: Output of Transparent Latch changes in response to the data input only when the Latch is 'Enabled'.
- Changes in input are straight away visible at the output

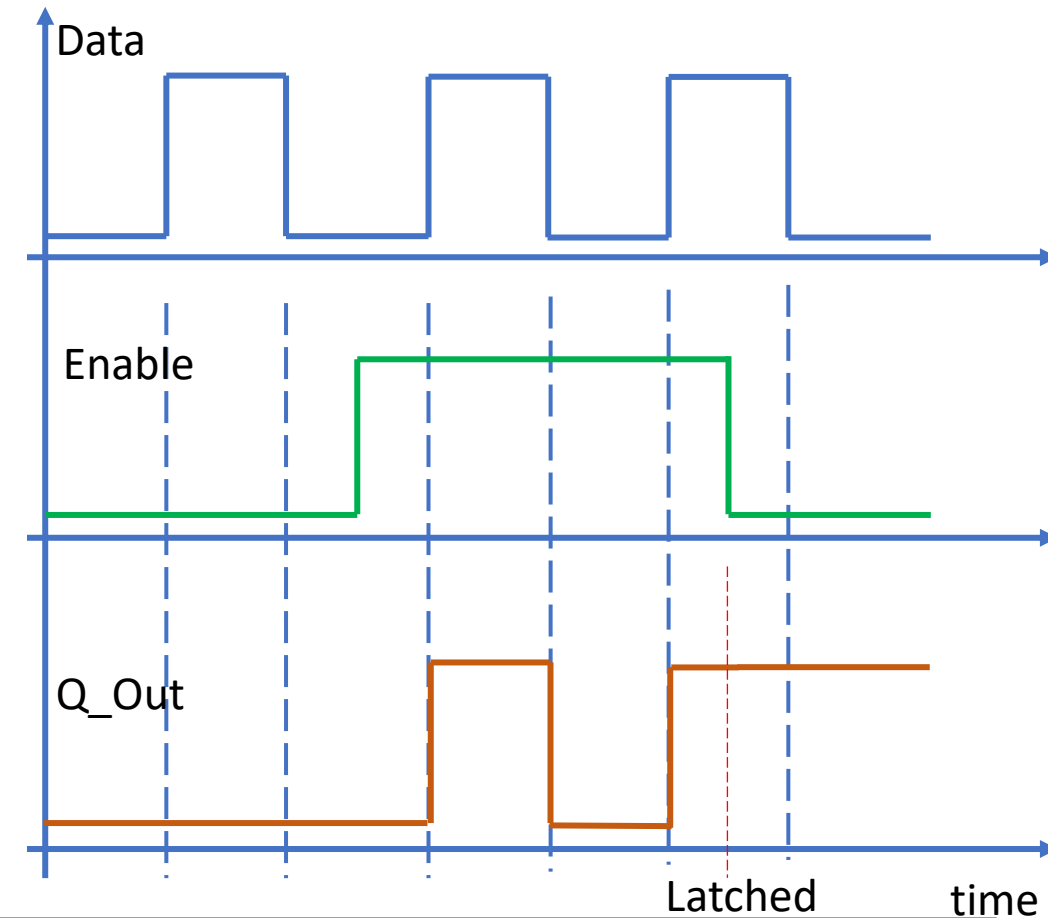
Transparent Latch Circuit

Timing Diagram – With Input Inverter



SR Latch With Enable (Without Inverter)

En	S	R	Next State of Q
0	X	X	No Change
1	0	0	No Change
1	0	1	Q=0; Reset State
1	1	0	Q=1; Set State
1	1	1	Not Allowed



When Enable is '0'; U='1' and V='1'; Hold Condition in SR Latch

Only these two states are valid when inverter is inserted